

D592

Environmental Health

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A state toxicologist, working from a lab that recently earned national accreditation, reviews rat studies to see how the rate of liver damage changes as the amount of a solvent fed to the animals goes up. The agency plans to release its findings at a spring press event. Which step of risk assessment is she performing?

- A. Risk communication.
- B. Dose-response assessment.
- C. Exposure assessment.
- D. Hazard identification.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An analyst surveys households to learn how many liters of well water residents drink each day and how long they have lived there. Which risk assessment step is this?

- A. Dose-response assessment.
- B. Risk characterization.
- C. Exposure assessment.
- D. Hazard identification, because all data on the well is hazard data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A county has a finished risk estimate for a leaking fuel tank near a creek. The creek is popular with anglers and has a small public boat launch. Officials must now choose between a \$2 million cleanup, a \$300,000 fence and signage plan, or monitoring only. What kind of decision is this?

- A. Risk management, because it weighs options, costs, and values after the science is done.
- B. Risk characterization, because it is the last step of risk assessment.
- C. Recreational use assessment, because anglers use the creek.
- D. Hazard identification, because the tank is leaking.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a two-year animal study published in a peer-reviewed toxicology journal, a pesticide showed no adverse effect at 5 mg/kg/day and liver changes at 20 mg/kg/day. Assessors apply an uncertainty factor of 10 for animal-to-human differences and 10 for variation among people. What is the reference dose?

- A. 0.5 mg/kg/day
- B. 0.2 mg/kg/day
- C. 0.05 mg/kg/day
- D. 500 mg/kg/day

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A chemical has a NOAEL of 3 mg/kg/day and a published reference dose of 0.003 mg/kg/day. What total uncertainty factor was applied?

- A. 100
- B. 3
- C. 1,000
- D. 0.001

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Testing finds a solvent in groundwater under an old dry cleaner that closed after 40 years in business. Every home in the area is on a municipal water system drawn from a reservoir 30 miles away, and there are no wells or basements near the plume. What is the best conclusion about exposure from drinking water?

- A. Residents are exposed because the solvent is present.
- B. Exposure is high because the business operated for 40 years.
- C. The solvent can never pose any risk to anyone.
- D. The drinking-water pathway is incomplete, so there is no current exposure by that route.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A neighborhood well contains a solvent at 0.3 mg/L. A 15 kg preschooler drinks 1 L a day; his 70 kg father drinks 2 L a day. The family moved in last spring and has a vegetable garden. What is the child's daily dose?

- A. 4.5 mg/kg/day
- B. 0.3 mg/kg/day
- C. 0.0086 mg/kg/day
- D. 0.02 mg/kg/day

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

An adult (70 kg, 2 L a day) and a toddler (10 kg, 1 L a day) drink the same water. How many times larger is the toddler's dose per kilogram?

- A. 0.29 times
- B. 3.5 times
- C. The same, because the water is the same.
- D. 7 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A furniture shop with 25 employees and a strong safety record uses a solvent-based finish that releases vapors. The owner, who plans to expand next year, is choosing a fix. Which option gives the most reliable protection?

- A. Rotate workers so no one finishes for more than two hours.
- B. Install local exhaust ventilation at each finishing station.
- C. Give each worker a respirator and fit test.
- D. Switch to a water-based finish that does not release the solvent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which of these controls for silica dust sits lowest on the hierarchy of controls?

- A. Scheduling cutting when fewer workers are nearby.
- B. Requiring workers to wear respirators.
- C. Buying precut materials so no cutting happens on site.
- D. Using saws with built-in water spray.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Early studies suggest a new lawn chemical may harm frog development and possibly human hormones, but results are mixed. The maker has sold it in three states since 2019. A city council asks what the precautionary principle supports. What is the best answer?

- A.** Limit or delay use until the maker shows it is safe.
- B.** Allow use because it has been sold since 2019 without a recall.
- C.** Ban every chemical with any uncertainty forever.
- D.** Allow use until harm is conclusively proven.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A state plans a fourth waste transfer station in a low-income neighborhood that already hosts three, plus a rail yard. The site is close to a highway exit, and the agency posted a legal notice in the newspaper. Residents say they were never asked. Which response best reflects environmental justice?

- A.** Assess cumulative impacts and hold accessible hearings where residents can shape the decision.
- B.** Proceed, because the highway access lowers truck traffic on local streets.
- C.** The legal notice already satisfied community involvement.
- D.** Residents chose to live near industry, so burden is not a fairness issue.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A state toxicologist, ~~working from a lab that recently earned national accreditation,~~ reviews rat studies to see how the rate of liver damage changes as the amount of a solvent fed to the animals goes up. ~~The agency plans to release its findings at a spring press event.~~ Which step of risk assessment is she performing?

Target word: name the step **Decides:** Relating amount given to the rate of effect is dose-response.

A	Camouflage	Answers the press-event detail; communication is not one of the four assessment steps.
B	Correct	It describes how the likelihood or severity of effect changes with dose.
C	Confusion set	Exposure assessment estimates how much people actually contact, not effects in animals.
D	Adjacent category	Hazard ID asks whether the solvent can cause harm at all, not how harm scales with amount.

2. Answer C Concept 1 • transfer

An analyst surveys households to learn how many liters of well water residents drink each day and how long they have lived there. Which risk assessment step is this?

Target word: name the step from the task **Decides:** Estimating how much and how long people contact the medium is exposure assessment.

A	Confusion set	Dose-response links dose to effect; it does not measure what residents drink.
B	Adjacent category	Characterization combines exposure with dose-response; it does not collect intake data.
C	Correct	Intake amount, frequency, and duration define exposure.
D	Absolute language	Not every study of a site is hazard ID; this one measures human contact.

3. Answer A Concept 2

A county has a finished risk estimate for a leaking fuel tank near a creek. ~~The creek is popular with anglers and has a small public boat launch.~~ Officials must now choose between a \$2 million cleanup, a \$300,000 fence and signage plan, or monitoring only. What kind of decision is this?

Target word: classify the decision **Decides:** Choosing among actions that differ in cost and feasibility is risk management.

A	Correct	Management uses the assessment plus cost, law, and values to pick an action.
B	Adjacent category	Characterization summarizes the risk estimate; it does not choose actions.
C	Camouflage	The anglers detail shapes exposure, but choosing among costed options is still management.
D	Confusion set	The hazard is already known; the question is what to do about it.

4. Answer C Concept 3

In a two-year animal study ~~published in a peer-reviewed toxicology journal~~, a pesticide showed no adverse effect at 5 mg/kg/day and liver changes at 20 mg/kg/day. Assessors apply an uncertainty factor of 10 for animal-to-human differences and 10 for variation among people. What is the reference dose?

Target word: reference dose **Decides:** $RfD = NOAEL / (10 \times 10)$.

A	Dropped element	Applies only one factor of 10.
B	Wrong stem number	Divides the LOAEL (20) instead of the NOAEL.
C	Correct	$5 / 100 = 0.05$ mg/kg/day.
D	Inverted operation	Multiplies by the uncertainty factors instead of dividing.

5. Answer C Concept 3 • transfer

A chemical has a NOAEL of 3 mg/kg/day and a published reference dose of 0.003 mg/kg/day. What total uncertainty factor was applied?

Target word: solve for the uncertainty factor **Decides:** $UF = NOAEL / RfD$.

A	Dropped element	Leaves out one factor of 10.
B	Wrong stem number	The NOAEL, not the factor.
C	Correct	$3 / 0.003 = 1,000$.
D	Inverted operation	Divides the RfD by the NOAEL.

6. Answer D Concept 4

Testing finds a solvent in groundwater under an old dry cleaner ~~that closed after 40 years in business~~. Every home in the area is on a municipal water system drawn from a reservoir 30 miles away, and there are no wells or basements near the plume. What is the best conclusion about exposure from drinking water?

Target word: is there a complete pathway **Decides:** No one drinks the groundwater, so the pathway lacks a point of exposure and route.

A	Dropped element	Presence is only the source and medium; the rest of the pathway is missing.
B	Camouflage	Years of operation shape how much contamination exists, not whether people contact it.
C	Absolute language	Other pathways (vapor, future wells) could become complete; only this pathway is broken now.
D	Correct	Without a point of exposure and a receptor, contamination alone does not create exposure.

7. Answer D Concept 5

A neighborhood well contains a solvent at 0.3 mg/L. A 15 kg preschooler drinks 1 L a day; his 70 kg father drinks 2 L a day. ~~The family moved in last spring and has a vegetable garden.~~ What is the child's daily dose?

Target word: child's dose in mg/kg/day **Decides:** $Dose = 0.3 \text{ mg/L} \times 1 \text{ L} / 15 \text{ kg}$.

A	Inverted operation	Multiplies by body weight instead of dividing.
B	Dropped element	The concentration, without dividing by body weight.
C	Confusion set	The father's dose ($0.6 / 70$), not the child's.
D	Correct	$0.3 \times 1 / 15 = 0.02$ mg/kg/day.

8. Answer B Concept 5 • transfer

An adult (70 kg, 2 L a day) and a toddler (10 kg, 1 L a day) drink the same water. How many times larger is the toddler's dose per kilogram?

Target word: ratio of doses **Decides:** Toddler 0.1 L/kg vs. adult 0.0286 L/kg.

A	Inverted operation	Adult over toddler, the reverse ratio.
B	Correct	$(1/10) / (2/70) = 0.1 / 0.0286 = 3.5$.
C	Adjacent category	Same concentration does not mean the same dose per kilogram.
D	Dropped element	Compares body weights only and ignores the adult's larger intake.

9. Answer D Concept 6

A furniture shop ~~with 25 employees and a strong safety record~~ uses a solvent-based finish that releases vapors. The owner, ~~who plans to expand next year~~, is choosing a fix. Which option gives the most reliable protection?

Target word: most protective control **Decides:** Substitution removes the hazard at the source; it ranks above engineering, administrative, and PPE.

A	Confusion set	An administrative control, which relies on people following rules.
B	Adjacent category	Engineering control is strong but ranks below substitution.
C	Inverted operation	PPE is the last resort, not the first.
D	Correct	Substitution ranks above every other option listed.

10. Answer B Concept 6 • transfer

Which of these controls for silica dust sits lowest on the hierarchy of controls?

Target word: least protective control **Decides:** PPE is the bottom rung.

A	Adjacent category	Administrative, one rung above PPE.
B	Correct	PPE depends on correct use by each worker every time.
C	Confusion set	Elimination, the top of the hierarchy.
D	Inverted operation	An engineering control, higher on the hierarchy.

11. Answer A Concept 7

Early studies suggest a new lawn chemical may harm frog development and possibly human hormones, but results are mixed. ~~The maker has sold it in three states since 2019.~~ A city council asks what the precautionary principle supports. What is the best answer?

Target word: application of precaution **Decides:** Plausible serious harm plus uncertainty supports protective action with the burden on the maker.

A	Correct	Precaution allows action before full proof and shifts the burden of proof.
B	Camouflage	Time on the market does not prove safety.
C	Absolute language	Precaution is proportionate, not a permanent ban on all uncertainty.
D	Inverted operation	That is the opposite burden of proof.

12. Answer A Concept 8

A state plans a fourth waste transfer station in a low-income neighborhood that already hosts three, plus a rail yard. ~~The site is close to a highway exit,~~ and the agency posted a legal notice in the newspaper. Residents say they were never asked. Which response best reflects environmental justice?

Target word: EJ response **Decides:** Cumulative burden on one community plus a lack of meaningful involvement.

A	Correct	Fair treatment and meaningful involvement are the two pillars of EJ.
B	Camouflage	Highway access is a logistics fact, not a response to the cumulative burden.
C	Dropped element	A notice is not meaningful involvement; residents must be able to influence the outcome.
D	Adjacent category	Frames a structural pattern as individual choice.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The four steps of risk assessment	Risk assessment runs hazard identification, dose-response assessment, exposure assessment, then risk characterization.	Treating any study of the chemical as hazard identification, or merging exposure with dose-response.	An analyst is doing one specific task and the question asks which step it is.
2	Risk assessment vs. risk management	Assessment is the scientific estimate of risk; management is the decision about what to do, weighing cost, feasibility, law, and public values.	Believing the risk number by itself decides the action.	A choice among options that differ in cost, feasibility, or politics.
3	Thresholds and the reference dose	For non-cancer effects, reference dose = NOAEL divided by the product of uncertainty factors.	Multiplying by uncertainty factors, or using the LOAEL when a NOAEL is given.	A NOAEL and one or more factors of 10 in the same stem.
4	Complete exposure pathway	Exposure needs a source, a medium, a point of exposure, a route into the body, and a receptor; break any link and there is no exposure.	Assuming the presence of a contaminant equals a health risk.	A contaminant is found, but something about people's contact with it is described.
5	Susceptible populations and dose per body weight	Dose = intake × concentration / body weight, so children, with more intake per kilogram, often get a larger dose than adults.	Assuming the same water or air gives everyone the same dose.	Body weights and intake rates for a child and an adult.
6	Hierarchy of controls	Elimination beats substitution, which beats engineering controls, then administrative controls, with PPE as the last resort.	Reaching for masks, gloves, or training first because they are quick and cheap.	Several control options listed and the question asks which is most (or least) protective.
7	Precautionary principle	When an activity raises a plausible threat of serious harm, protective action can be taken before cause and effect is fully proven, and the proponent carries the burden of showing safety.	Waiting for conclusive proof of harm before acting.	Scientific uncertainty plus a potentially serious or irreversible harm.
8	Environmental justice	No group should bear a disproportionate share of environmental burdens, and affected communities deserve meaningful involvement in decisions.	Explaining unequal exposure as the result of individual choices, or equating a public notice with meaningful involvement.	Facilities or pollution clustered in a low-income or minority neighborhood.

Course scope (public description): This course provides learners with the foundational knowledge and relationship people have with their environment, the risk management choices made, and the resulting associations that affect health and physical well-being for the individual, communities, and susceptible populations.